CS-104 Retrieval Quiz (ANSWER KEY)

Fall 2026 · Week 4, Wednesday

Name: _____

1.

What does this code print?

```
x = 95
if x >= 90:
    print("You've got an A")
if x >= 80:
    print("You've got a B")
else:
    print("Keep studying!")
```

a) Only `You've got an A`

b) Only `You've got a B`

c) `You've got an A` **and then** `You've got a B` **[correct]**

d) `You've got an A` and then `Keep studying!`

2.

Which expression is **not** equivalent to `13 <= age < 20`?

a) `13 <= age and age < 20`

b) `age >= 13 and age < 20`

c) `not (age < 13 or age >= 20)`

d) `13 < age <= 20` **[correct]**

3.

What does this code print?

```
a = 5
result = (a == 3 or 4)
if result:
    print("yes")
else:
    print("no")
```

a) `yes`**, because** `False or 4` **gives** `4`**, which is truthy [correct]**

b) `no`, because a is not 3

c) Nothing; it raises a `TypeError`

d) Nothing; neither branch runs

4.

Write a code segment that reads a pH value (a float) from the user and prints `error` if it is outside the range 0 to 14. Otherwise it prints `acid` if the pH is below 7, `base` if it is above 7, and `neutral` if it is exactly 7.

Sample solution:

```
ph = float(input("pH: "))
if ph < 0 or ph > 14:
    print("error")
elif ph < 7:
    print("acid")
elif ph > 7:
    print("base")
else:
    print("neutral")
```